Mission Configuration

Start Position

Start X

0

Start Y

0

Goal Position

Goal X

9

Goal Y

7

Obstacle coordinates

2,0; 2,1; 2,2

Planning Mode

Choose how to plan

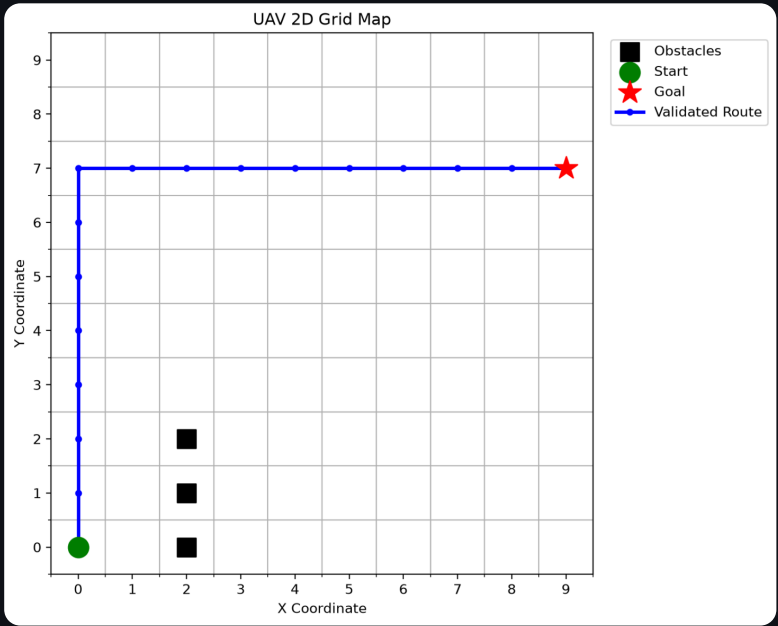
- Fast Mode (A* + Safety)
- AI-Assisted Mode (Microsoft Agent Framework)

Plan and Validate Route

UAV Path Planning Using Agentic AI

Hybrid architecture: deterministic A* path planning, independent safety validation, and optional AI assistance.

AI-assisted route independently validated. Path passed all safety checks.



Route Results

Grid Size	Route Waypoints	Total Moves
10 × 10	17	16

Direction Changes	Horizontal Moves	Vertical Moves
1	9	7

Obstacle collisions: 0 — no obstacle collisions detected.

Selected validated route:

```
(0, 0) → (0, 1) → (0, 2) → (0, 3) → (0, 4) →
```

Alternative Route Analysis

Found 3 valid route(s). Best returned route: 16 moves and 1 direction changes. Search stopped before exhausting all candidates; additional routes may exist.

Fewest Moves Among Can…	Fewest Turns at That Move…
16	1

Candidate Route 1

Moves	Waypoints	Direction Changes
16	17	1

Matches the best move-count and direction-change metrics among the validated candidates returned.

```
(0, 0) → (0, 1) → (0, 2) → (0, 3) → (0,
```

Candidate Route 2

Moves	Waypoints	Direction Changes
16	17	2

Ties for the fewest moves found, but has more direction changes.

(0, 0) → (0, 1) → (0, 2) → (0, 3) → (0,

Candidate Route 3

Moves	Waypoints	Direction Changes
16	17	2

Ties for the fewest moves found, but has more direction changes.

(0, 0) → (0, 1) → (0, 2) → (0, 3) → (0,

The search stopped before exhausting its queue. Additional routes may exist. Metrics describe only the validated candidates returned by the search.

Route Explanation

Selected route: 16 moves and 17 waypoints. Direction changes: 1. Horizontal moves: 9; vertical moves: 7. Obstacle collisions: 0. Manhattan lower bound: 16 moves. The selected route reaches the Manhattan lower bound, so it is a shortest possible route for four-direction unit-cost movement. Alternative analysis returned 3 route(s). The best move count among those routes is 16; the fewest direction changes among routes with that move count is 1. The selected route matches the best move-count and direction-change metrics among the returned routes. The alternative-route search stopped before exhausting its search queue. Additional routes may exist, so the returned candidates are not an exhaustive comparison.

View AI Agent's Mission Interpretation